

Chapter 1: Introduction to Python

Python is a high-level, interpreted, interactive, and object-oriented scripting language. Created by Guido van Rossum in the late 1980s, Python's design philosophy emphasizes code readability with the use of significant indentation. Key Features of Python: 1. Easy to Learn and Read: Python has straightforward, English-like syntax that makes it easy for beginners to grasp. 2. Interpreted Language: Code is executed line by line, making debugging straightforward and interactive programming possible. 3. Dynamically Typed: You do not need to explicitly declare variable data types prior to assignment; Python determines types automatically at runtime. 4. Portable and Cross-Platform: Python scripts run seamlessly across Windows, macOS, Linux, and Unix environments. 5. Extensive Standard Libraries and Packages: Support for third-party tools via pip expands its capabilities into data science, web development, and AI.